

PID

$$G(s) = K \left(1 + \frac{1}{T_I s} + T_D s \right) = \frac{K}{s} \left(T_D s^2 + s + \frac{1}{T_I} \right)$$

$$= \frac{K}{s} \left(T_D s^2 + \left(1 + \frac{T_D}{T_I} \right) s + \frac{1}{T_I} \right)$$

$$= \frac{K}{s} (T_D s + 1) \left(s + \frac{1}{T_I} \right)$$

$$\underline{\underline{\frac{K (T_D s + 1) (s + \frac{1}{T_I})}{s (T_D s + 1)}}}$$

